PG No. 25 C_{6v} $6mm$ [hexagonal]

Table 1: Representation matrices

Irrep.						
A_1	$1:\begin{bmatrix}1\end{bmatrix}$	$3^+_{001}:\begin{bmatrix}1\end{bmatrix}$	$3^-_{001}:\begin{bmatrix}1\end{bmatrix}$	$2_{001}:\begin{bmatrix}1\end{bmatrix}$	$6^-_{001}:\begin{bmatrix}1\end{bmatrix}$	$6^+_{001}:\begin{bmatrix}1\end{bmatrix}$
	$\mathrm{m}_{110}:\begin{bmatrix}1\end{bmatrix}$	$\mathrm{m}_{100}:\begin{bmatrix}1\end{bmatrix}$	$\mathrm{m}_{010}:\begin{bmatrix}1\end{bmatrix}$	$\mathrm{m}_{1-10}:\begin{bmatrix}1\end{bmatrix}$	$\mathrm{m}_{120}:\begin{bmatrix}1\end{bmatrix}$	$\mathrm{m}_{210}:\begin{bmatrix}1\end{bmatrix}$
A_2	$1:\begin{bmatrix}1\end{bmatrix}$	$3^+_{001}:\begin{bmatrix}1\end{bmatrix}$	$3^-_{001}:\begin{bmatrix}1\end{bmatrix}$	$2_{001}:\begin{bmatrix}1\end{bmatrix}$	$6^-_{001}:\begin{bmatrix}1\end{bmatrix}$	$6^+_{001}:\begin{bmatrix}1\end{bmatrix}$
	$\mathrm{m}_{110}:\begin{bmatrix}-1\end{bmatrix}$	$\mathrm{m}_{100}:\begin{bmatrix}-1\end{bmatrix}$	$\mathrm{m}_{010}:\begin{bmatrix}-1\end{bmatrix}$	$\mathrm{m}_{1-10}:\begin{bmatrix}-1\end{bmatrix}$	$\mathrm{m}_{120}:\begin{bmatrix}-1\end{bmatrix}$	$\mathrm{m}_{210}:\begin{bmatrix}-1\end{bmatrix}$
B_1	$1:\begin{bmatrix}1\end{bmatrix}$	$3^+_{001}:\begin{bmatrix}1\end{bmatrix}$	$3^-_{001}:\begin{bmatrix}1\end{bmatrix}$	$2_{001}:\begin{bmatrix}-1\end{bmatrix}$	$6^-_{001}:\begin{bmatrix}-1\end{bmatrix}$	$6^+_{001}:\begin{bmatrix}-1\end{bmatrix}$
	$\mathrm{m}_{110}:\begin{bmatrix}1\end{bmatrix}$	$\mathrm{m}_{100}:\begin{bmatrix}1\end{bmatrix}$	$\mathrm{m}_{010}:\begin{bmatrix}1\end{bmatrix}$	$\mathrm{m}_{1-10}:\begin{bmatrix}-1\end{bmatrix}$	$\mathrm{m}_{120}:\begin{bmatrix}-1\end{bmatrix}$	$\mathrm{m}_{210}:\begin{bmatrix}-1\end{bmatrix}$
B_2	$1:\begin{bmatrix}1\end{bmatrix}$	$3^+_{001}:\begin{bmatrix}1\end{bmatrix}$	$3^-_{001}:\begin{bmatrix}1\end{bmatrix}$	$2_{001}:\begin{bmatrix}-1\end{bmatrix}$	$6^-_{001}:\begin{bmatrix}-1\end{bmatrix}$	$6^+_{001}:\begin{bmatrix}-1\end{bmatrix}$
	$\mathrm{m}_{110}:\begin{bmatrix}-1\end{bmatrix}$	$\mathrm{m}_{100}:\begin{bmatrix}-1\end{bmatrix}$	$\mathrm{m}_{010}:\begin{bmatrix}-1\end{bmatrix}$	$\mathrm{m}_{1-10}:\begin{bmatrix}1\end{bmatrix}$	$\mathrm{m}_{120}:\begin{bmatrix}1\end{bmatrix}$	$\mathrm{m}_{210}:\begin{bmatrix}1\end{bmatrix}$
E_1	$1:\begin{bmatrix}1&0\\0&1\end{bmatrix}$	$3^+_{001}:\begin{bmatrix}-\frac{1}{2}&-\frac{\sqrt{3}}{2}\\\frac{\sqrt{3}}{2}&-\frac{1}{2}\end{bmatrix}$	$3^-_{001}:\begin{bmatrix}-\frac{1}{2}&\frac{\sqrt{3}}{2}\\-\frac{\sqrt{3}}{2}&-\frac{1}{2}\end{bmatrix}$	$2_{001}:\begin{bmatrix}-1&0\\0&-1\end{bmatrix}$	$6^-_{001}:\begin{bmatrix}\frac{1}{2}&\frac{\sqrt{3}}{2}\\-\frac{\sqrt{3}}{2}&\frac{1}{2}\end{bmatrix}$	$6^+_{001}:\begin{bmatrix}\frac{1}{2}&-\frac{\sqrt{3}}{2}\\\frac{\sqrt{3}}{2}&\frac{1}{2}\end{bmatrix}$
	$\mathrm{m}_{110}:\begin{bmatrix}\frac{1}{2}&-\frac{\sqrt{3}}{2}\\-\frac{\sqrt{3}}{2}&-\frac{1}{2}\end{bmatrix}$	$\mathrm{m}_{100}:\begin{bmatrix}-1&0\\0&1\end{bmatrix}$	$\mathrm{m}_{010}:\begin{bmatrix}\frac{1}{2}&\frac{\sqrt{3}}{2}\\\frac{\sqrt{3}}{2}&-\frac{1}{2}\end{bmatrix}$	$\mathrm{m}_{1-10}:\begin{bmatrix}-\frac{1}{2}&\frac{\sqrt{3}}{2}\\\frac{\sqrt{3}}{2}&\frac{1}{2}\end{bmatrix}$	$\mathrm{m}_{120}:\begin{bmatrix}1&0\\0&-1\end{bmatrix}$	$\mathrm{m}_{210}:\begin{bmatrix}-\frac{1}{2}&-\frac{\sqrt{3}}{2}\\-\frac{\sqrt{3}}{2}&\frac{1}{2}\end{bmatrix}$
E_2	$1:\begin{bmatrix}1&0\\0&1\end{bmatrix}$	$3^+_{001}:\begin{bmatrix}-\frac{1}{2}&-\frac{\sqrt{3}}{2}\\\frac{\sqrt{3}}{2}&-\frac{1}{2}\end{bmatrix}$	$3^-_{001}:\begin{bmatrix}-\frac{1}{2}&\frac{\sqrt{3}}{2}\\-\frac{\sqrt{3}}{2}&-\frac{1}{2}\end{bmatrix}$	$2_{001}:\begin{bmatrix}1&0\\0&1\end{bmatrix}$	$6^-_{001}:\begin{bmatrix}-\frac{1}{2}&-\frac{\sqrt{3}}{2}\\\frac{\sqrt{3}}{2}&-\frac{1}{2}\end{bmatrix}$	$6^+_{001}:\begin{bmatrix}-\frac{1}{2}&\frac{\sqrt{3}}{2}\\-\frac{\sqrt{3}}{2}&-\frac{1}{2}\end{bmatrix}$
	$\mathrm{m}_{110}:\begin{bmatrix}-\frac{1}{2}&\frac{\sqrt{3}}{2}\\\frac{\sqrt{3}}{2}&\frac{1}{2}\end{bmatrix}$	$\mathrm{m}_{100}:\begin{bmatrix}1&0\\0&-1\end{bmatrix}$	$\mathrm{m}_{010}:\begin{bmatrix}-\frac{1}{2}&-\frac{\sqrt{3}}{2}\\-\frac{\sqrt{3}}{2}&\frac{1}{2}\end{bmatrix}$	$\mathrm{m}_{1-10}:\begin{bmatrix}-\frac{1}{2}&\frac{\sqrt{3}}{2}\\\frac{\sqrt{3}}{2}&\frac{1}{2}\end{bmatrix}$	$\mathrm{m}_{120}:\begin{bmatrix}1&0\\0&-1\end{bmatrix}$	$\mathrm{m}_{210}:\begin{bmatrix}-\frac{1}{2}&-\frac{\sqrt{3}}{2}\\-\frac{\sqrt{3}}{2}&\frac{1}{2}\end{bmatrix}$